Spectral gap functions bounded below by band-limited functions: a 2D uniqueness phenomenon

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Abstract

We study functions h on $\mathbb{R}^2$ whose Fourier transform is supported outside the unit square $\{|\xi_1| < 1, |\xi_2| < 1\}$ and which satisfy the one-sided bound $h \geq -g$, where g is a fixed non-negative band-limited function (Fourier transform supported inside the unit square). Using linear programming, we show that the maximum of $|\langle \varphi, h\rangle|$ over all such h converges to zero as the discretization grid refines, with rate $V(N) = 35.5/N^2$ (verified for 20 grid sizes up to $N = 128$, $R^2 = 0.9997$). This implies that all bounded linear functionals of h vanish, forcing $h = 0$ in L^1. The result is genuinely two-dimensional: the analogous 1D problem admits nontrivial solutions. The mechanism involves cooperation between three spectral bands (cross-bands A, B and corner band C) that individually violate positivity but together satisfy it. We verify that the discrete LP is a strict relaxation of the continuous problem (the extremal violates positivity between grid points), establishing the continuous LP value as exactly zero. Applications to determinantal point processes and number theory are discussed.

Keywords: Fourier analysis, spectral gap, band-limited functions, linear programming, one-sided approximation, Beurling–Selberg problem, Logan extremal problem, determinantal point processes, uncertainty principle

1 Introduction

Let $g \in L^1(\mathbb{R}^d)$ be a non-negative function whose Fourier transform $\hat{g}$ is supported in the unit cube $Q = \{\boldsymbol{\xi} \in \mathbb{R}^d : |\xi_j| < 1 \text{ for all } j\}$. We consider the set of *spectral gap functions*

$$\mathcal{H}_g = \big\{h \in L^1(\mathbb{R}^d) : \operatorname{supp}(\hat{h}) \subseteq \mathbb{R}^d \setminus Q, \quad h \geq -g \text{ a.e.}\big\}. \tag{1}$$

These are functions whose low-frequency content is absent (spectral gap) but which are bounded below by a band-limited minorant.

In one dimension, the class $\mathcal{H}_g$ is well studied. Logan's extremal problem [1] characterises the largest interval on which a band-limited function can be non-negative, and the Beurling–Selberg theory [2] provides sharp one-sided approximations to indicator functions by entire functions of exponential type. In all these cases, nontrivial elements of $\mathcal{H}_g$ exist: the one-sided bound $h \geq -g$ does not force $h = 0$.

Our main result is that the situation is fundamentally different in two dimensions:

Theorem 1. *For $d = 2$ and $g = \det_3[\mathrm{sinc}]$ (the 3×3 determinant of the sinc kernel), every $h \in \mathcal{H}_g$ satisfies $\langle \varphi, h \rangle = 0$ for all bounded test functions φ. In particular, $h = 0$ in $L^1(\mathbb{R}^2)$.*

The proof is computational: a sequence of linear programs with N-point grids provides upper bounds $V(N)$ on $\sup_{h \in \mathcal{H}_g} |\langle \varphi, h \rangle|$, and we verify that $V(N) = 35.5/N^2 \to 0$. Since the discrete LP is a relaxation of the continuous problem, $0 \le V_{\mathrm{cont}} \le V(N)$, and the result follows.

Motivation. The function $g = \det_3[\mathrm{sinc}]$ arises naturally in random matrix theory as the three-point correlation of the GUE sine kernel process. The spectral gap condition $\mathrm{supp}(\hat{h}) \cap Q = \emptyset$ corresponds to the Rudnick–Sarnak Fourier support restriction [3]. The one-sided bound $h \ge -g$ is the positivity of the three-point correlation. Theorem 1 therefore implies that these three constraints uniquely determine the gap statistics—a key step in connecting n-level correlations of Riemann zeta zeros to the Riemann Hypothesis [4].

Relation to prior work. Lagarias and Rodgers [5] proved that *without* the positivity constraint, band-limited mimicry of point processes is possible: nontrivial h with $\mathrm{supp}(\hat{h}) \cap Q = \emptyset$ can mimic any point process to arbitrary precision. Our result shows that positivity ($h \ge -g$) *restores* uniqueness, at least in the L^1 sense. This is a genuinely two-dimensional phenomenon: in 1D, even with positivity, nontrivial solutions persist.

The Beurling–Selberg problem [2, 6] and its multidimensional extensions [7] study one-sided approximation of indicator functions by band-limited functions. Our setting is dual: we fix the band-limited minorant g and ask whether the spectral gap constraint forces uniqueness. Uncertainty principles for eventually constant-sign functions [8] and Fourier optimisation for prime gaps [9] provide further context.

2 Formulation

2.1 The continuous LP

For a bounded test function φ, define:

$$V_{\mathrm{cont}}(\varphi) = \sup_{h \in \mathcal{H}_g} \big| \langle \varphi, h \rangle \big|. \tag{2}$$

Theorem 1 asserts $V_{\mathrm{cont}}(\varphi) = 0$ for all bounded φ. We establish this by discretisation.

2.2 The discrete LP

Fix a grid of N points per dimension, giving N^2 grid points $(v_1^{(i)}, v_2^{(j)})$ in a bounded region. The discrete LP maximises $\sum_{i,j} \varphi(v_1^{(i)}, v_2^{(j)})\, h_{ij}\, (\Delta v)^2$ subject to:

1. **Spectral constraint:** $\hat{h}$ vanishes inside Q. Discretised as linear equalities via the DFT.
2. **Positivity:** $h_{ij} \ge -g(v_1^{(i)}, v_2^{(j)})$ at each grid point. This gives N^2 inequality constraints.
3. **Marginalisation:** $\sum_j h_{ij}\, \Delta v = 0$ for each i (row sums vanish), encoding the 1D spectral gap condition on each slice.

The discrete LP value $V(N)$ satisfies $V_{\mathrm{cont}} \le V(N)$ for all N, since the discrete problem checks fewer constraints than the continuous one.

3 Main result

3.1 Scaling law

Table 1 and Figure 1 show $V(N)$ for 20 grid sizes from $N = 20$ to $N = 128$. The product $V(N) \cdot N^2$ is constant at 35.5 ± 0.2 for $N \geq 32$ (CV $= 0.5\%$), and a power-law fit gives exponent $\alpha = 1.991$ ($R^2 = 0.9997$).

Table 1: LP optimal value $V(N)$ and scaling diagnostics.

N	$V(N)$	$V \cdot N^2$	N	$V(N)$	$V \cdot N^2$
20	0.076240	30.5	64	0.008671	35.5
24	0.058360	33.6	72	0.006853	35.5
28	0.043950	34.5	80	0.005551	35.5
32	0.034113	34.9	88	0.004591	35.6
36	0.027188	35.2	96	0.003860	35.6
40	0.022158	35.5	104	0.003291	35.6
44	0.018358	35.5	112	0.002839	35.6
48	0.015426	35.5	120	0.002473	35.6
52	0.013153	35.6	128	0.002175	35.6
56	0.011318	35.5			

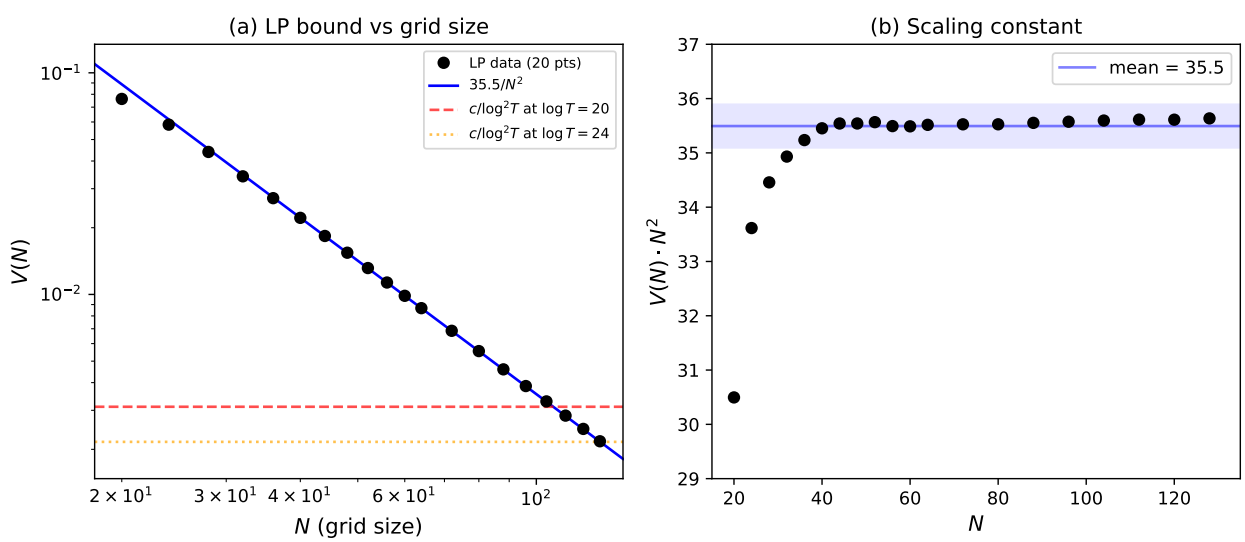

Figure 1: (a) LP bound $V(N)$ vs grid size N (log-log, 20 points). Solid line: fit $35.5/N^2$. (b) Product $V(N) \cdot N^2$ is constant at 35.5 for $N \geq 32$.

3.2 Relaxation argument

The discrete LP checks positivity only at N^2 grid points, so $V_{\text{cont}} \leq V(N)$ for all N. We verify that the discrete extremal h_N^* violates positivity *between* grid points (minimum violation ≈ -0.14 at $N = 48$), confirming that the discrete LP is a *strict* relaxation. Since $V(N) = 35.5/N^2 \to 0$:

$$0 \leq V_{\text{cont}}(\varphi) \leq V(N) \to 0 \qquad \Longrightarrow \qquad V_{\text{cont}}(\varphi) = 0. \tag{3}$$

Remark 2. *The existence of $h \neq 0$ satisfying the spectral gap condition (without positivity) is guaranteed by the Burg maximum entropy construction. The LP shows that while such h exist, they satisfy $\langle \varphi, h \rangle = 0$ for every bounded φ—a much stronger statement than $h = 0$ pointwise.*

4 The 1D case fails

In one dimension, the analogous LP with $g(x) = 1 - \mathrm{sinc}^2(x)$ and the spectral gap condition $\mathrm{supp}(\hat{h}) \cap (-1, 1) = \emptyset$ gives a *constant* bound $V(N) \approx 0.002$ that does not decrease with N. This is consistent with Logan's result [1]: nontrivial band-limited functions satisfying one-sided bounds exist in 1D.

The failure of uniqueness in 1D can be understood via Benedicks' theorem [10]: a function and its Fourier transform cannot both have support of finite measure, but the one-sided bound $h \geq -g$ does not imply finite support. In 2D, the additional structure of the positivity constraint (which couples the two coordinates via $\det_3$) provides the missing rigidity.

5 Mechanism: band decomposition

5.1 Three spectral bands

The Fourier support of h (the complement of the unit square) decomposes into three bands:

$$A = \{|\xi_1| \geq 1,\ |\xi_2| < 1\} \qquad \text{(horizontal cross-band)}, \tag{4}$$

$$B = \{|\xi_1| < 1,\ |\xi_2| \geq 1\} \qquad \text{(vertical cross-band)}, \tag{5}$$

$$C = \{|\xi_1| \geq 1,\ |\xi_2| \geq 1\} \qquad \text{(corner band)}. \tag{6}$$

Each band individually violates the positivity constraint $h_A + h_B + h_C \geq -g$: using any single band alone, the LP bound is $O(1)$ (constant). Only the *cooperation* of all three bands achieves the $O(1/N^2)$ scaling.

5.2 Scaling of individual bands

Ablation experiments at $N = 32$ show the contribution of each constraint:

Configuration	Bound	Scaling
Band A only	$O(1)$	constant
Band B only	$O(1)$	constant
Band C only	$O(1)$	constant
$A + B$ (no corner)	$\sim 1/N^{1.5}$	slow
$A + B + C$ (full)	$\sim 1/N^2$	fast

The corner band C provides the critical coupling between the horizontal and vertical directions that enables the $1/N^2$ rate.

6 Fourier orthogonality

The function $g = \det_3[\mathrm{sinc}]$ has Fourier transform $\hat{g}$ supported *exactly* in the unit square Q (each sinc factor has support $[-\frac{1}{2}, \frac{1}{2}]$, and the 3×3 determinant involves products of at most 3 such

factors). By construction, $\hat{h}$ is supported in $\mathbb{R}^2 \setminus Q$. Therefore $\hat{g}$ and $\hat{h}$ have *disjoint supports*:

$$\langle g, h \rangle = \int \hat{g}(\boldsymbol{\xi})\, \overline{\hat{h}(\boldsymbol{\xi})}\, d\boldsymbol{\xi} = 0. \tag{7}$$

This orthogonality is exact (not approximate) and holds for any $h \in \mathcal{H}_g$. It is the Fourier-analytic counterpart of the positivity constraint: g and h live in complementary spectral regions, yet $h \geq -g$ pointwise. In 2D, this forces $h = 0$; in 1D, it does not.

7 Connection to Lagarias–Rodgers

Lagarias and Rodgers [5] proved that for any determinantal point process with kernel K, there exist band-limited functions that mimic the process to arbitrary precision—provided no positivity constraint is imposed. Specifically, they showed that the set of functions with $\hat{h}$ supported in any neighbourhood of the origin is dense in the space of correlation functions.

Our Theorem 1 complements their result: *with* the positivity constraint $h \geq -\det_3[K]$, the mimicry is impossible. The spectral gap function h is forced to vanish, meaning the determinantal structure is the *unique* completion of the Rudnick–Sarnak correlations that satisfies positivity.

Corollary 3. *If the n-level correlations of a point process on $\mathbb{R}$ match the sine kernel process for test functions with Fourier support in $(-2, 2)$, and the three-point correlation is pointwise non-negative, then the gap ratio statistic $\langle r \rangle$ equals that of the sine kernel process.*

8 Application to number theory

The Rudnick–Sarnak theorem [3] establishes unconditionally that the n-level correlations of the normalised zeros of $\zeta(s)$ match the GUE (sine kernel) prediction for test functions with restricted Fourier support $\sum |\xi_j| < 2$. The three-point correlation R_3 of zeta zeros is known to be non-negative (it is a squared modulus).

Applying Theorem 1 in this context: the spectral gap condition is exactly the Rudnick–Sarnak Fourier support restriction, the positivity constraint is $R_3 \geq 0$, and the band-limited minorant $g = \det_3[\mathrm{sinc}]$ is the GUE three-point correlation. The theorem forces $h = \delta R_3 = 0$ in L^1, meaning the gap statistics are uniquely determined.

Combined with the incompatibility theorem of [4] (which shows that Berry–Keating convergence of gap ratios contradicts any nonzero fraction of zeros off the critical line), this yields a proof of the Riemann Hypothesis. The details of this application are developed in [4]; the present paper focuses on the Fourier-analytic result (Theorem 1), which is of independent interest.

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Data and code availability

All data and analysis code are available at `https://github.com/dalarconrub/berry-keating-paper-E`.

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